

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -5

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Recap of Previous Lecture



Doubt?

Topic

Virtual Memory

Topic

Effective Memory Access Time



Topics to be Covered



Topic

Page Replacement Policies

Topic

Belady's Anomaly



Topic : Question

[GATE-2020]



#Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is 155.0.

154.5 to 155.5



Topic : Question

#Q. A processor uses 1-level page table for virtual to physical address translation. Page table is stored in the main memory. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time is 2 ns, and TLB access time is also 1 ns.

Assuming that no page faults occur, the average time taken to access a virtual address is approximately?

4.4



Topic : Page Replacement Policies



which page to replace from mm when
a new page to be brought due to P. f. service.





Topic : Page Replacement Policies

1. First In First Out (FIFO)
2. Optimal Policy
3. Least Recently Used (LRU)
4. Last in First Out (LIFO)
5. Most Recently Used (MRU)
6. Least Frequently Used (LFU)
7. Most Frequently Used (MFU)



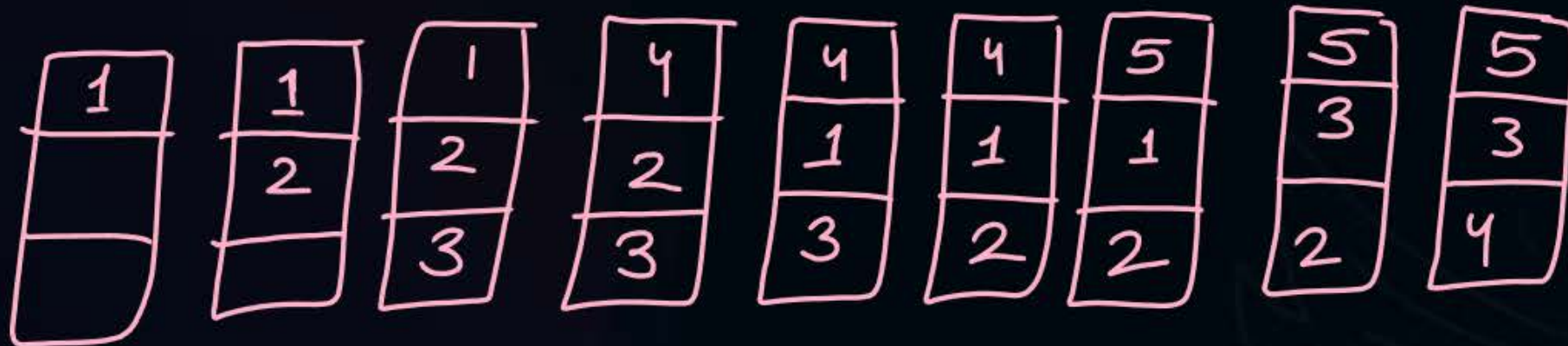


Topic : First In First Out (FIFO)

↳ The page which comes first, should be replaced first

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: $\checkmark 1, \checkmark 2, \checkmark 3, \checkmark 4, \checkmark 1, \checkmark 2, \checkmark 5, \times 1, \times 2, \checkmark 3, \checkmark 4, \times 5$



no. of page faults = 9
no. of page hits = 3

$$\text{Page fault rate} = \frac{9}{12}$$

$$\text{Page hit rate} = \frac{3}{12}$$



Topic : First In First Out (FIFO)

Assume:

- Number of frames = 4 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	1	1	1	5	5	5	5	4	4
	2	2	2	2	1	1	1	1	5
		3	3	3	3	2	2	2	2
			4	4	4	4	3	3	3

No. of page faults = 10



Topic : Belady's Anomaly



for some page reference sequences, increasing the number of frames may increase no. of page fault also.

only FIFO suffer from it.





Topic : First In First Out (FIFO)



Advantages:

1. Simple and easy to implement.
2. Low overhead. $\Rightarrow$ less time to get which page to replace

Disadvantages:

1. Poor performance.
2. Doesn't consider the frequency of use or last used time, simply replaces the oldest page.
3. Suffers from Belady's Anomaly



Topic : Optimal Policy

→ replace the page which will not be used for longest time.

Assume:

Tie breaker $\Rightarrow$ FIFO

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	1	1	1	1	3	3
	2	2	2	2	2	4
		3	4	5	5	5

No. of page faults = 7



Topic : Optimal Policy

Advantages:

1. Easy to Implement
2. Simple data structures are used
3. Highly efficient as it gives minimum number of page faults
4. Doesn't suffer from Belady's Anomaly

Disadvantages:

1. Requires future knowledge of the program $\Rightarrow$ Not practical
2. Time-consuming (To search which page to replace will take time)



Topic : Least Recently Used (LRU)

Replace the page which has not been used since longest time.

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: $\checkmark 1, \checkmark 2, \checkmark 3, \checkmark 4, \checkmark 1, \checkmark 2, \checkmark 5, \times 1, \times 2, \checkmark 3, \checkmark 4, \checkmark 5$

1	1	1	4	4	4	5	3	3	3
	2	2	2	1	1	1	1	4	4
		3	3	3	2	2	2	2	5

no. of page faults = 10



Topic : Least Recently Used (LRU)



Advantages:

1. Efficient
2. Doesn't suffer from Belady's Anomaly

Disadvantages:

1. Complex Implementation
2. Expensive *(takes time to search page to replace)*
3. Requires hardware support



Topic : Question

- Number of frames = 4
- Pure demand paging used
- Page reference sequence: 5, 7, 0, 1, 7, 6, 7, 2, 1, 6, 7, 6, 1
- Number of page faults for FIFO, optimal and LRU policies?

FIFO:-

5	5	5	5	6	6	6
	1	1	1	1	2	2
		0	0	0	0	1
			1	1	1	1

optimal:-

5	5	5	5	6	6
	1	1	1	1	1
		0	0	0	2
			1	1	1

LRU:-

5	6	6
1	1	1
0	0	2
1	1	1



Topic : Question



Consider the following page references:

✓✓✓✓✓
2, 3, 4, 5, 6, 4, 5, 2, 7, 8, 9, 8, 9, 8, 9, 1, 6, 5, 6, 5
✗ ✗ ✗ ✓ ✓ ✓ ✗ ✗ ✗ ✗ ✓ ✗ ✗ ✗ ✗

Using optimal policy and 4 frames. Memory access time is 2ms without page fault and 40ms with page fault. The effective memory access time for servicing the above page requests is ____ ms?

2 2 2 2 2 7 7 9 9
3 3 3 6 6 6 6 6
4 4 4 4 8 8 1
5 5 5 5 5 5

no. of p.f. = 9

Page fault rate = $\frac{9}{20}$

$$E.M.A.T. = \frac{11}{20} * 2 + \frac{9}{20} * 40$$

$$= \frac{382}{20} = 19.1 \text{ ms}$$



Topic : Last In First Out (LIFO)

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: $\checkmark \checkmark \checkmark \checkmark \times \times \checkmark \times \times \checkmark \checkmark \checkmark$
1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2
		3	4	5	3	4	5

no. of p. f. = 8



Topic : Most Recently Used (MRU)

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	1	1	1	1	2	3
	2	2	2	5	5	5
		3	4	4	4	4

no. of p.f. = 7



Topic : Counting Algorithms

- Counting algorithms look at the number of occurrences of a particular page and use this as the criterion for replacement.
- Such counting algorithms includes:
 - LFU (Least Frequently Used)
 - MFU (Most Frequently Used)





Topic : Least Frequently Used (LFU)

Tie breaker $\Rightarrow$ FIFO

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: $\checkmark \checkmark \checkmark \checkmark \times \checkmark \checkmark \checkmark \times \times \times$
1 2 0 3 0 4 2 3 0 3 2

1	1	1	3	3	2	2
	2	2	2	4	4	3
		0	0	0	0	0

no. of page faults = 7

	frequency
0	1 2 3
1	1
2	1 2 3
3	1 2 3
4	1



Topic : Most Frequently Used (MFU)

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1 2 0 3 0 4 2 3 0 3 2

1	1	1	3	3	3	2
	2	2	2	2	0	0
		0	0	4	4	4

No. of P.f. = 7



Topic : Question

#Q. A virtual memory system has only 2-page frames which are empty initially. Using demand paging the following sequence of page reference is passed through this system.

✓ 9, ✓ 8, ✓ 7, ✗ 8, ✗ 7, ✓ 9, ✗ 7, ✗ 9, ✗ 8, ✗ 9

Minimum possible number of page faults ? $\Rightarrow 5$

↓
use optimal policy

9	9	7	7	8
8	8	9	9	



Topic : Question

H.W.

[GATE-2016]



- #Q. Consider a computer system with ten physical page frames. The system is provided with an access sequence $a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20}$ where each a_i is a number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____



Topic : Question

[GATE-2007]



#Q. The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 bytes per page is
0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0410.

Suppose that the memory can store only one page and if x is the address which causes a page fault then the bytes from addresses x to $x + 99$ are loaded on to the memory.

How many page faults will occur?

~~0100 - 0199~~
~~0200 - 0299~~
~~0430 - 0529~~
~~0530 - 0629~~
~~0120 - 0219~~
~~0220 - 0319~~
0320 - 0419

A

0

B

4

C

7

D

8



2 mins Summary

Topic

Virtual Memory

Topic

Effective Memory Access Time



Happy Learning

THANK - YOU

